INCH-POUND

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MILITARY SPECIFICATION

THEODOLITES, SURVEYING, DIRECTIONAL; THIRD ORDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of night illuminated direction-type theodolites with accessories, cases, and tripods.

1.2 Classification. The theodolite shall be one of the following types, as specified (see 6.2):

Type I - 1 minute
Type II - 0.2 mils

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).
V-T-276	-	Thread, Cotton.
W-L-111	-	Lamp, Electric Incandescent Miniature, Tungsten-Filament.
TT-F-325	-	Filler, Engraving, Stamped Marking.
CCC-C-419	-	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-D-950	-	Dyeing and After Treating Processes for Cotton Cloths.

MILITARY

MIL-B-18	-	Battery, Dry.
MIL-B-18/9	-	Battery, Dry, BA-30.
MIL-V-173	-	Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronics and Associated Equipment).
MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-H-639	-	Hooks (for Belts, Straps and Equipage Items).
MIL-C-675	-	Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-T-3530	-	Thread and Twine; Mildew Resistant or Water Repellent Treated.
MIL-S-3534	-	Surveying Instruments and Accessories, Packaging of.
MIL-T-14091	-	Tripod, Surveying: Extension-Leg-Type.
MIL-R-14328	-	Rubber Sheet Synthetic Medium Soft, General Purpose Gasket Material (for Extreme Climate Condition).
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-43566	-	Tape, Textile, Cotton, General Purpose, Natural or in Colors.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standard for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the specification.

NATIONAL BUREAU OF STANDARDS (NBS)

Special Publication 374 - Method for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rules for the use of Volatile Organic Compounds (VOC).

(Applications for copies should be addressed to the South Coast Air Quality Management District, Attn: Public Affairs Department, 9150 Flair Drive, El Monte, CA 91731).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The theodolites shall be optical-reading, direction-type instruments with a least circle graduation of 1 minute (0.2 mils), and shall be equipped with a repeating clamp. Each theodolite shall weigh not more than 15.4 pounds (7.0 kilograms) and shall consist essentially of an alidade with an integral illumination system and a tribrach. Associated equipment shall include a carrying case, a transport case, a tool and accessory case, a

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theodolite cover, a compass, and, when specified herein, a telescope level, a knapsack, winterization kit, and an autocollimation eyepiece.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The theodolite shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the theodolite may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the theodolite may be newly fabricated from recovered materials to the maximum extent practicable, provided the theodolite produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the theodolite.

3.4 Environmental conditions.

3.4.1 Operating temperature. The theodolite, tripod, cases, and accessories shall operate as specified herein and shall not be damaged when subjected to ambient temperatures from 125 ^{0}F (51.7 ^{0}C) to -40 ^{0}F (-40 ^{0}C).

3.4.2 Storage temperature. The theodolite, cases, and accessories shall not be damaged by storage under the following conditions:

Air temperature:

High - 155 ^{0}F. (68.3 ^{0}C) Time period - 4 hours.
Low - -50 ^{0}F. (-45.6 ^{0}C) Time period - 6 hours. No solar radiation.

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3.4.3 Solar radiation. The theodolite shall operate as specified herein and shall not be damaged by full impact of solar radiation of 355 British thermal units per square foot per hour (Btu/ft^2/hr) at an air temperature of 125 ^{0}F (51.7 ^{0}C) for 4 hours.

3.4.4 Rainfall. The theodolite, carrying case, and transport case shall show no signs of water penetration when subjected to rainfall at a rate of 3 inches (7 cm), ±1 inch (±2.5 cm), per hour, as follows:

Theodolite - 5 minutes.
Carrying case - 1 hour.
Transport case - 1 hour.

3.4.5 Sand and dust. When the theodolite and carrying case are subjected to blowing sand and dust, the theodolite shall not be damaged, shall not show sand or dust penetration in the telescope, optical system, or on bearings, and shall not show lens abrasion. The carrying case shall not show sand or dust penetration.

3.4.6 Vibration. The theodolite and the carrying case shall not be damaged by vibration when tested as specified in 4.6.2.18.

3.4.7 Shock. The theodolite and cases shall not be damaged by shock when tested as specified in 4.6.2.19.

3.4.8 Humidity. The item shall not be damaged and shall operate within the accuracy specified herein when subjected to relative humidity of at least 95 percent at air temperature varying between -25 ^{0} F (-31.7 ^{0}C) and 110 ^{0}F (43.3 ^{0}C).

3.5 Alidade assembly. The alidade assembly shall contain the leveling device, telescope, horizontal and vertical circles, plate and altitude level, standards, bearings, and provision for electric illumination. Provisions shall be made on the alidade for attaching a compass.

3.5.1 Telescope. The telescope shall show an erecting image, shall be internal focusing, and shall consist of an optical system, eyepieces, sights, and a retractable or removable sunshade. The overall length of the telescope shall be not greater than 7 inches (180 mm) when the diopter is set to zero, and shall transit at the eyepiece end when the eyepiece is fully turned out. The telescope shall balance when focused at infinity with the sunshade in the extended position. The interior of the telescope shall contain no noticeable internal reflections. The change in the line of collimation, resulting from the travel of the negative lens over the focusing range, shall not change the line of near-to-infinity horizontal and vertical focus-collimation of more than 12 seconds (0.06 mils) over a focusing range of from approximately 15 feet (4.6 m) to infinity.

3.5.1.1 Optical system. The optical properties shall conform to the following requirements:

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a. Magnification, telescope with standard eyepiece - 28 diameters, ±3 diameters.
b. Effective aperture of objective, minimum - 1.42 inches (36 mm).
c. Field o3 1, 4^G^U
. Focus, minimum, not greater than - 7 feet (2.2 m).
d. Resolution - 4 seconds of arc or less.

Except for the reticle, the internal lens-air surfaces within the telescope shall be uniformly coated with a reflection-reducing film in accordance with MIL-C-675. The field of view shall be substantially flat from the center to a zone outside the stadia wires equal to one-fourth of the stadia interval and shall show no marked curvature from there to the edge of the field. The principal axis of the lens system shall be in alignment and shall be coincident with the collimation of the telescope in adjustment.

3.5.1.2 Reticle. The reticle shall consist of a pattern of opaque lines similar to that shown in figure 1 registered on optical glass (see 6.4). The crosslines shall extend to the edge of the field of view. The single vertical crossline shall subtend equal angles with the adjacent double lines within ±0.5 second. The width of the lines shall be 2.5 to 4.0 microns. The double vertical lines shall subtend an apparent angle of 40 second, +5 or -10 seconds. The radius of the solar circle shall be 15 minutes 30 seconds, ±10 seconds. The horizontal and vertical stadia lines shall be spaced so that 1/100 of the distance from a stadia rod at a known distance greater than 100 feet (30.5 m) from the instrument shall be subtended with an error in reading not to exceed 0.3 percent. The lines shall subtend 5 to 7 minutes of arc in length and shall be parallel to the corresponding crosslines. The reticle shall be mounted in a metal cell which shall not become distorted during adjustment of the collimation. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors shall be not more than 0.2 minute (0.06 mils).

3.5.1.3 Telescope eyepiece. The eyepiece shall be constructed so that the eye of an operator can be placed in a position that will coincide with the exit pupil. The eyepiece shall be equipped with a diopter scale marked from zero in plus and minus units and shall have an index mark referencing the setting. The telescope eyepiece shall be designed for placing of the eyepiece prism (see 3.5.1.3.2), and interchange with the autocollimation eyepiece (see 3.5.1.3.4).

3.5.1.3.1 Telescope eyepiece filter. The telescope eyepiece filter for solar observations shall be attachable without the use of tools. The telescope eyepiece filter shall contain a black or Nichrome-coated or equal neutral filter with a density range of between 5.0 and 6.0. If a Nichrome-coated or equal filter is furnished, the coated surface shall be toward the eyepiece lens.

3.5.1.3.2 Telescope eyepiece prism. The telescope eyepiece prism shall permit favorable positioning for viewing without disturbing the focus of the

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telescope eyepiece. The prism shall permit observation of ground points and angles within 26 degrees from the zenith with the telescope both in the direct and reversed position. The reticle pattern shall be clearly visible and the magnification/change shall not exceed 2 percent for any position of the prism.

3.5.1.3.3 Eyepiece prism filter. The eyepiece prism filter shall contain an absorption-type black or Nichrome-coated or equal neutral filter having a density range between 5.0 and 6.0. The Nichrome-coated or equal surface shall be placed adjacent to the telescope eyepiece.

3.5.1.3.4 Autocollimation eyepiece. When specified (see 6.2), an autocollimation eyepiece shall be furnished. This eyepiece shall be interchangeable with the normal eyepiece without the use of tools. The magnification of the telescope with the eyepiece shall be not less than 18 diameters.

3.5.2 Standards. The standards shall be adjustable for height or they shall be factory adjusted and set permanently. The horizontal axis of the telescope shall be perpendicular to the vertical axis of the theodolite so that the error due to the difference in the height of the standards shall be not more than 12 seconds (0.06 mils) of arc. When required for mounting the compass yoke, the standards shall have the necessary receptacles.

3.5.3 Optical plummet. The optical plummet shall be integral with the alidade and shall be adjustable. The reticle of the optical plummet shall have a circular or cross pattern placed so that it will produce an image concentric with the vertical axis of the instrument. The reticle pattern shall enclose the image of a hub tack at tripod height and a 4-inch diameter target at 110 feet (102 mm at 33 meters). The optical systems hall produce a clear image throughout the operating range from tripod height to 110 feet (33 meters).

3.5.4 Circles. Horizontal and vertical circles and breakdown scales shall be optical glass. Graduations and numbers shall be uniform, legible, and permanent. The graduations on the circles shall be viewed in the theodolite with transmitted light.

3.5.4.1 Horizontal circle. The horizontal circle shall be equipped with a horizontal motion clamp, tangent screw, and repeating clamp to set the circle to any predetermined reading. Engaging and releasing the repeating clamp shall not cause more than a 0.1 minute change in the circle reading. The reading shall not change when the alidade is rotated with the repeating clamp engaged.

3.5.4.2 Eccentricity. The maximum eccentricity of the circles shall not exceed 0.33 minute (0.1 mils).

3.5.4.3 Vertical circle. The vertical circle shall be equipped with a vertical tangent screw, clamp, and level tangent screw (where applicable).

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3.5.4.4 Circle identification. The images of the horizontal and vertical circles in the microscope shall be identified by the initial letters "H" or "Hz" and "V", respectively, adjacent to the correct circle image.

3.5.4.5 Breakdown scale. The breakdown scale shall be fixed and superimposed on the images of segments of the horizontal and vertical circles. A difference between any two consecutive circle graduations and the scale graduation lines shall not exceed 0.2 minute (0.06 mils).

3.5.5 Microscope. A microscope shall be provided through which the horizontal and vertical circles and breakdown scales shall be viewed. The eyepiece of the microscope shall be at the side of the telescope eyepiece, and shall rotate with it or shall be rotatable through 180 degrees when mounted on the standard adjacent to and in the same horizontal plane with the telescope. The reading section of both the horizontal and vertical circles and both breakdown scales shall be visible simultaneously through the reading microscope. The images of both horizontal and vertical circles, index lines, and breakdown scales shall appear erect and in the same focal plane.

3.5.5.1 Eyepiece prism. When the reading microscope is attached to the telescope and cannot be read in every position of the telescope, an eyepiece prism for the microscope shall be furnished. The eyepiece prism shall give an erect image of the reading scales. The use of the eyepiece prism shall not change the magnification of the microscope.

3.5.6 Levels. The instrument shall be equipped with a plate level, circular level, self-indexing device or an altitude level where applicable.

3.5.6.1 Plate level. The plate level shall be graduated to a 2 mm spacing and shall have a sensitivity of 30 seconds, ±8 seconds, for each 2 mm. Graduations shall be filled with a paste conforming to TT-F-325, type II, color black, or by fused ceramic markings applied to the outer surface of the glass. The levels shall be mounted as close as practicable to the vertical axis and shall be protected. The level shall be provided with an external or internal adjustment by the operator from the outside of the instrument without disassembly of the instrument.

3.5.6.2 Circular level. The instrument shall be equipped with a circular level. The circular level shall have a sensitivity of 10 minutes, ±3 minutes, for each 2 mm.

3.5.6.3 Vertical circle indexing. The indexing of the vertical circle shall be controlled by a self-indexing device, except that when specified (see 6.2), an altitude level shall be furnished. When the vertical circle in the direct position reads 90 degree, or 1600 mils, the telescope shall be horizontal.

3.5.6.3.1 Self-indexing device. The self-indexing device shall be damped so that oscillation is stopped in not more than 5 seconds of time. The device shall be accurate within 0.2 minute of arc or 0.06 mils, as applicable.

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3.5.6.3.2 Altitude level. The altitude level vial for indexing the vertical circle shall have a sensitivity of 30 seconds, ±5 seconds, for each 2 mm and shall be ungraduated. The level shall be read by the coincidence method through a train of prisms and a rotatable viewing glass on the vertical circle standard. The image of the coincident bubble ends shall appear in the viewing glass at the operating temperature specified herein. A tangent screw shall be provided for bringing the bubble ends into coincidence in the viewing device. The level shall be provided with an external or internal adjustment. The internal adjustment shall be accessible for field adjustment by the operator from the outside of the instrument without disassembly of the instrument.

3.6 Tribrach. The tribrach shall consist of the leveling device, base plate, and a device for locking the alidade assembly. The base plate shall have at its center a 5/8 inch nut of the unified coarse thread series (5/8-11-UNC-2B), class 2B, 11 threads per inch, as defined in FED-STD-H28. The leveling device shall permit leveling of the theodolite when the base plate is 6 degrees out of level.

3.6.1 Interchangeability. Tribrachs shall be interchangeable with all alidades procured under this specification from the same manufacturer. Figure 2 shows a stub that has been found acceptable for this use (see 6.4).

3.6.2 Maintenance of centering. The construction of the tribrach shall be such that removing and returning the alidade shall cause a change in the centering of the alidade not to exceed 0.002 inch (0.05 mm).

3.7 Bearings. The vertical axis shall be supported on ball-type bearings. The trunnion axis shall be supported on ball-type or plain bearings.

3.8 Compass. Unless otherwise specified (see 6.2), the theodolite shall be equipped with an induction-damped, circular magnetic compass containing a magnetic needle balanced on a replaceable pivot. The caging mechanism shall lock the needle from outside the compass. The compass shall be compensated for the theodolite at the factory with provision for field adjustment for individual theodolites and shall have counterbalances for leveling. The compass shall be capable of setting off the declination from 0 to 180 degree (3200 mils) east and west. With the magnetic declination set at zero, the focusing eyepiece shall be located above the telescope eyepiece, and the index line of the compass shall be parallel to the instrument line of sight. The compass shall be securely attached to the standards of the theodolite, and the telescope shall transit when the compass is in place. The compass, when securely fastened to the theodolite, shall permit carrying of the theodolite mounted on its tripod without danger of the compass loosening from the theodolite. The compass shall be installed and removed without the use of tools. The compass shall have an accuracy of ±30 minutes (9 mils) of arc. The compass dial shall rotate uniformly without sticking. Two spare pivots shall be furnished with each compass (see 3.18).

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3.9 Housings. Metal housings shall be provided for screw threads for the principal parts of the theodolite, circles, bearings, and reticles. Screwcaps shall be furnished for exposed attachment threads.

3.10 Operating screw. Where applicable, leveling screws and tangent screws shall have a take-up for wear adjustment, and operating screws shall have heads constructed for recognition in darkness. Tangent screws shall be removable for cleaning. The altitude-level tangent screw head and the horizontal- and vertical- motion tangent screw heads shall be distinctly different in shape, where applicable.

3.11 Theodolite cover. A lightweight, flexible, waterproof cover shall be furnished to protect the theodolite and compass from damage from all environmental conditions specified herein.

3.12 Illumination. The theodolite shall be illuminated by natural and electrical means. Natural illumination shall be mirrors. Electrical illumination shall be supplied by a lighting attachment with batteries.

3.12.1 Natural lighting. One or two adjustable mirrors coated on the second surface with reflective material shall illuminate the vertical and horizontal circles and their corresponding breakdown scales. Moving or adjusting the mirror shall not cause a variation in the circle reading of more than 0.10 minute (0.03 mils) of arc.

3.12.2 Electrical lighting. The lighting attachment(s) shall illuminate the horizontal and vertical circles, breakdown scales, and, if applicable, the plate level and altitude level. The reticle shall be provided with a means for varying the intensity of light. Lighting attachments shall have frosted glass covers or frosted bulbs conforming to W-L-111. When the theodolite is operated in the dark, there shall be no visible light beyond 100 feet (30.5 meters) except that within the telescope field of view. A rubber gasket conforming to MIL-R-14328 shall be provided along the edges of the lighting attachment, if necessary to prevent the escape of light. The battery receptacles shall be clearly marked to show polarity.

3.12.2.1 Wiring. If the wiring is used for the lighting equipment, sufficient slack shall be allowed for field repair of broken connections, provided the slack does not in any way impede or interfere with the operation of the theodolite. The wire shall be extra flexible, shall show no evidence of damage, and shall not crack at -25 ^{0}F (-31.7 ^{0}C) when wrapped around a 3/8 inch (9 mm) diameter mandrel.

3.12.2.2 Hand lamp. When specified (see 6.2), a hand lamp shall be provided and shall connect to a receptacle on the battery case. The hand lamp shall have a switch and a connecting cord not less than 5 feet long (1.52 meters). The socket of the hand lamp shall be threaded with the miniature lamp holder thread defined in FED-STD-H28. The cord shall meet the requirements of 3.12.2.1.

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3.12.2.3 Bulbs. Bulbs, if used in the lighting attachment or in the hand lamp, shall conform to W-L-111.

3.12.2.4 Battery and lighting accessory case. When specified (see 6.2), a battery and lighting accessory case of metal, glass fiber reinforced plastic laminate, or a high impact plastic that will withstand dropping without breakage or damage under the low temperature specified in 3.4.1 shall be furnished. The case shall contain space for four batteries conforming to MIL-B-18 and MIL-B-18/9. The batteries shall not be furnished. The batteries shall be connected in a series parallel arrangement to provide 3 volts. The case shall be provided with two receptacles. One receptacle shall be equipped with a rheostat with a marked "OFF" position. The other receptacle shall provide full voltage. The interior of the case shall provide space for the hand lamp, the connector cords, and the lighting attachment(s) for night illumination (see 3.18); when applicable, four replacement lamps shall be furnished and space provided for them in the interior of the case. The case shall have provisions for mounting on the tripod.

3.12.2.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.12.2.5.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.2.5.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with the rules associated with volatile organic compounds (VOC) as established by local air quality rulings and the South Coast Air Quality Management District rulings.

3.13 Tripod. When specified (see 6.2), an extension-leg tripod conforming to MIL-T-14091 shall be furnished.

3.14 Carrying case. A carrying case of metal, glass fiber reinforced plastic laminate, or high impact molded plastic shall protect the theodolite against damage. Each case shall contain a desiccant adequate to the volumes

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of the container and provision for storage of the adjusting pins and jeweler's screwdriver. The glass fiber reinforced plastic laminate cases shall have reinforced corners. The theodolite shall be fastened within the case to prevent the theodolite from falling out when the case is opened. The case shall be provided with heavy-duty catches for closing. If the glass fiber reinforced plastic laminate case contains a filler of plastic foam and skin or other material, the filler shall be removable for cleaning. The weight of the carrying case shall not exceed 13 pounds (5.9 kg).

3.15 Transport case. A reusable transport case that will house and protect the theodolite in its carrying case against rainfall, shock, and vibration during field transport, shipment, and storage shall be furnished. The transport case shall not be damaged, shall protect the theodolite in its carrying case from damage, and shall not allow water to enter the case when tested as specified in 4.6.2.14.3, 4.6.2.18.2, and 4.6.2.19.2. The case shall be made of molded plastic, reinforced plastic laminate, or polyethylene and shall weigh not more than 15 pounds (6.8 kg). The case shall have removable inserts contoured to fit snugly around the carrying case. The case shall have carrying handle(s) and locks(s), or clasp(s). The glass fiber reinforced plastic laminate case shall have metal reinforced corners. All hardware shall be of corrosion-resistant metal or treated to resist corrosion.

3.16 Winterization kit. The winterization kit shall include covers for all control knobs and leveling screws, shields for the telescope, the microscope, and the optical plummet eyepieces. The kit shall be contained in a cotton duck carrying case.

3.16.1 Case. The case shall be of cotton duck and shall contain all the components of the winterization kit. The case shall have a slide fastener, a belt loop, and double belt hook conforming to MIL-H-639, type I, class 1. The cotton duck used shall conform to CCC-C-419, type I, No. 12, hard-texture and rendered fungus-resistant and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f, and dyed olive drab 7. Webbing used shall conform to MIL-W-530, type IIa, class 4, dyed olive 7. Tape shall conform to MIL-T-43566, type I, class 4. Thread used in fabricating the case shall conform to V-T-276, type III B, ticket No. 10.4, dyed olive drab 7, and rendered fungus-resistant in accordance with MIL-T-3530, type I, class 1.

3.17 Knapsack. When specified (see 6.2), a knapsack shall be furnished for backpacking the theodolite in its carrying case. The knapsack shall be fabricated from cotton duck conforming to CCC-C-419, type I, No. 12, hard texture and rendered fungus-resistant and water-repellent in accordance with CCC-D-950, type I, class B, inhibitor f. Webbing used shall conform to MIL-W-530, type IIa, class 4. Tape shall conform to MIL-T-43566, type I, class 4. Thread used in the fabrication of the knapsack shall conform to V-T-276, type III B, ticket No. 10/4, and shall be rendered fungus-resistant in accordance with MIL-T-3530, type I, class 1. The knapsack shall be fitted with pockets for carrying accessories and shall be large enough to allow for shrinkage. The cap shall fit snugly over the top of the knapsack to prevent

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penetration of rain. Shoulder straps shall be cushioned or padded. The color of the knapsack shall be as specified (see 6.2).

3.18 Tools and accessories. When specified (see 6.2), the following items shall be furnished with each theodolite:

Item	Quantity
Brush, lens, camel's hair.	1
Chamois skin, 8 x 8 inches (20 x 20 cm).	1
Compass, magnetic (see 3.8).	1
Orangewood sticks (commercial grade).	1 package
Lens tissue.	1 package
Container, lubricant, with not less than 1/4 ounce of grease conforming to MIL-G-23827 or equal (see 3.19).	1
Cover, Waterproof (see 3.11).	1
Prism, eyepiece, telescope (see 3.5.1.3.2).	1
Prism, eyepiece, microscope (see 3.5.5.1).	1
Filter, Solar (see 3.5.1.3.1).	1
Filter, prism eyepiece, telescope (see 3.5.1.3.3).	1
Lighting attachments (see 3.12.2).	1 set
Pins, adjusting.	2
Pivots for compass, spare (see 3.8).	2
Screwdriver, flat tip.	1
Screwdriver, jeweler's.	1

The contractor may add items to this list as required for field maintenance and adjustment. The adjusting pins and screwdrivers shall correctly fit the intended adjustments of the theodolites.

3.18.1 Tool and accessory case. The tools and accessories shall be contained in a cotton-webbing case, except the jeweler's screwdriver and the adjusting pins may be stored in the carrying case. The case shall be similar in construction to the case specified in 3.16.1 and shall have pockets or loops for the tools and accessories.

3.19 Lubrication. Unless otherwise specified (see 6.2), the theodolite shall be lubricated prior to assembly with grease conforming to MIL-G-23827 or equal. The grease shall permit the theodolite to operate throughout the specified temperature ranges.

3.20 Identification marking. Each theodolite, battery case, and accessory case shall be identified in accordance with MIL-STD-130. The marking of the theodolite shall be placed on one of the standards. Marking shall include the letters "US", the manufacturer's name or trademark, the serial number of the

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theodolite, the graduation type marking "DEG" or "MIL". Additional markings on the theodolite shall include the NSN assigned to the theodolite, the contract number under which the theodolite was procured, and the year of manufacture. The carrying case shall be marked with the letters "US", the type designation, as applicable, and the serial number of the theodolite. The markings specified for the carrying case shall be painted on the transport case. Canvas case and the knapsack, when specified, shall be identified by stenciling using ink conforming to A-A-208. The serial number of the theodolite with which it is used, the graduation type marking "DEG" or "MIL", as applicable, and the letters "US" shall be engraved on the compass leg.

3.21 Treatment and painting. The portions of the theodolite, cases, and accessories normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color shall be as specified (see 6.2). The interior of the telescope and the optical train shall be matte black.

3.22 Type I theodolite.

3.22.1 Horizontal circle. The horizontal circle shall be graduated at 1 degree intervals. Each degree graduation shall be numbered. The numbering shall begin with 0 degrees and shall be consecutive through 359 degrees both clockwise and counterclockwise. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 3 seconds of arc. The outside diameter of the graduated area shall be not less than 1.97 inches (50 mm).

3.22.1.1 Breakdown scale. The breakdown scale shall be fixed, shall be 60 minutes long, and shall be graduated at 1 minute intervals. The graduations shall be numbered at 10 minute intervals. The numbering shall begin with 0 minutes and shall be consecutive through 60 minutes both clockwise and counterclockwise.

3.22.1.2 Numbering. The numbers shall be symmetrically oriented with respect to the graduations. A method shall be used to differentiate the clockwise and counterclockwise direction numbering. This may include slanting the numbering in opposite directions, providing arrows to indicate the direction the numbering increase, or different background shades for each numbering direction.

3.22.2 Vertical circle. The vertical circle shall be graduated at 1 degree intervals. Each degree graduation shall be numbered. The numbering shall be symmetrically oriented with respect to the graduations, and shall begin with 0 degrees and be consecutive through 359 degrees. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 3 seconds of arc. The outside diameter of the graduated area shall be not less than 1.97 inches (50 mm).

3.22.2.1 Breakdown scale. The breakdown scale shall be fixed, shall be 60 minutes long, and shall be graduated at 1 minute intervals. The graduations shall be numbered at 10 minute intervals. The numbering shall be

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symmetrically oriented with respect to the graduations and shall begin with 0 minutes and be consecutive through 60 minutes.

3.22.3 Telescope level. A telescope level shall be supplied with each type I theodolite. The level shall be of the coincidence type, and the level vial shall have a sensitivity of 60 seconds, ±10 seconds, for each 2 mm. The level shall be adjustable and removable.

3.22.4 Magnetic compass. The compass shall be graduated at 1 degree intervals and numbered every 10 degrees. The numbering shall be from 0 to 350, increasing from left to right as viewed in the reading eyepiece. Compass reading shall be zero when the theodolite is aimed north (magnetic). Accuracy of the compass shall be ±30 minutes.

3.22.5 Operating screws. The horizontal- and vertical-motion tangent screws shall rotate their respective axes not less than 10 minutes nor more than 25 minutes of arc per revolution. The horizontal- and vertical-motion tangent screws shall be long enough to produce at least 3 degree of motion over the entire run.

3.23 Type II theodolite.

3.23.1 Horizontal circle. The horizontal circle shall be graduated at 10 mils intervals. Each graduation shall be numbered. The numbering shall begin with 0 mils and shall be consecutive through 639 (0) mils, and shall increase as the alidade with telescope in the direct position is rotated clockwise. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 0.015 mils. The outside diameter of the graduated area shall be not less than 1.97 inches (50 mm).

3.23.1.1 Breakdown scale. The breakdown scale shall be fixed, shall be 10 mils long, and shall be graduated at 0.2 mils intervals. The graduations shall be numbered at 1 mil intervals. The numbering shall begin with 0 mils and shall be consecutive through 10 mils.

3.23.1.2 Numbering. The numbers shall be symmetrically oriented with respect to the graduations. Three digits shall be used for all numbers.

3.23.2 Vertical circle. The vertical circle shall be graduated at 10 mils intervals. Each graduation shall be numbered. The numbering shall be in accordance with 3.22.1.2, and shall begin with 0 mils and be consecutive through 639 (0) mils. The circle shall be graduated so that no error in spacing of the graduations, either individual or cumulative, shall exceed 0.015 mils. The outside diameter of the graduated area shall be not less than 1.97 inches (50 mm).

3.23.2.1 Breakdown scale. The breakdown scale shall be fixed, shall be 10 mils long and shall be graduated at 0.2 mils intervals. The graduations shall be numbered at 1 mil intervals. The outside diameter of the graduated area shall be not less than 1.97 inches (50 mm).

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Review activities:

3.23.3 Magnetic compass. The compass shall be graduated at 10 mils intervals and numbered every 100 mils. The numbering shall be from 0 to 63, increasing from left to right as viewed in the reading eyepiece. Compass reading shall be zero (0) when the theodolite is aimed north (magnetic). Accuracy of the compass shall be ±9 mils of arc.

3.23.4 Operating screws. The horizontal- and vertical-motion tangent screws shall rotate their respective axes not less than 3 mils nor more than 7.5 mils of arc per revolution. The horizontal- and vertical-motion tangent screws shall be long enough to produce at least 53.3 mils of motion over the entire run.

3.24 Workmanship. All threads requiring movement, and sliding and rotatable parts, shall move without binding, backlash, or end play. Clamping actions shall be positive and, when unclamped, shall not cause drag. Focusing components and assemblies shall retain their focus when finger pressure is released, and the telescope shall not change focus when transited. No lateral motion shall exist between the telescope and standards. Lenses shall be free of scratches, chips, air bubbles between cemented elements, dirt between the glass and coated film, or cloudiness. Optics shall show sharpness of focus and distinct images to provide favorable definition. Metal surfacings on the optical components for all prisms and lenses shall not shift. There shall be no flickering of the illumination systems when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Components and assemblies shall show no evidence of forced attachment. Bearing surfaces shall show no scratches or surface blemishes. Mirrors shall not be tarnished, and backing shall be uniform and unspotted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known

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defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Prior to assembly inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection of packaging (see 4.8).

4.3 Prior to assembly inspection.

4.3.1 Tests. Prior to assembly, the horizontal and vertical circles shall be tested for accuracy, the level vials shall be tested for sensitivity and the reticles shall be measured for reticle line weight as specified in 4.6.2.2.3. Failure of the applicable test shall be cause for rejection of the applicable items.

4.4 First article inspection.

4.4.1 Examination. The theodolite, carrying case, transport case, battery case, and accessories shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The theodolite, carrying case, transport case, battery case, and accessories shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined using MIL-STD-105, table I and table IIa.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the lot.

4.5.3 Tests.

4.5.3.1 Individual. Each theodolite shall be tested as specified in 4.6.2.3 through 4.6.2.16. Failure of any test shall be cause for rejection of the theodolite.

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4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be tested in accordance with the schedule shown in table I in the order shown. Failure of any test shall be cause for rejection of the lot.

4.6 Inspection procedure.

4.6.1 Examination. The theodolite, carrying case, transport case, battery case and accessories shall be examined as specified herein for the following defects:

Major

101. Weight of theodolite and case not as specified (see 3.1, 3.14, and 3.15).
102. Material not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of materials, materials finishes, or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the theodolite (see 3.3.4).
107. Alidade assembly not as specified (see 3.5).
108. Telescope will not transit and balance as specified (see 3.5.1).
109. Coated optics not as specified (see 3.5.1.1).
110. Optical system does not produce a sharp image throughout the entire focusing range and is not free from internal reflection, spherical and chromatic erration, coma, astigmatism, or distortion (see 3.5.1.1).
111. Reticle pattern not as specified (see 3.5.1.2).
112. Reticle lines not clean-cut, appear fuzzy, or do not appear opaque (see 3.5.1.2).
113. Eyepieces not as specified (see 3.5.1.3).
114. Solar filter not as specified (see 3.5.1.3.1).
115. Circles and breakdown scales not graduated as specified (see 3.5.4).
116. Graduations on circles and breakdown scales not clear and clean-cut or appear fuzzy (see 3.5.4).
117. Horizontal repeating clamp not as specified (see 3.5.4.1).
118. Eyepiece prisms not furnished or not as specified (see 3.5.5.1).
119. Tribrach not as specified (see 3.6).
120. Bearings not as specified (see 3.7).
121. Compass not as specified (see 3.8).
122. Housings not as specified (see 3.9).
123. Operating screws not as specified (see 3.10).
124. Theodolite cover not as specified (see 3.11).
125. Fungus and moisture treatment of canvas items not as specified (see 3.12.2.5).
126. Illumination system not as specified (see 3.12).
127. Battery and lighting accessory case not as specified (see 3.12.2.4).

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128. Cases not as specified (see 3.14, 3.15, and 3.16).
129. Knapsack not as specified (see 3.17).
130. Tools and accessories not as specified (see 3.18).
131. Lubrication not as specified (see 3.19).
132. Workmanship not as specified (see 3.24).

Minor

201. Identification marking not as specified (see 3.20).
202. Treatment and painting not as specified (see 3.21).

4.6.2 Tests. The theodolite, carrying case, transport case, battery case, and accessories shall be tested as specified herein in accordance with the schedule and sequence specified in table I. After adjustment of the theodolite by the contractor (see 4.6.2.1), readjustment may be made only at the conclusion of the shock, vibration, and extreme environment tests.

TABLE I. Test schedule.

Test	Paragraph	First Article	Requirement Paragraph	Quality Conformance
1	2	3	4	5
Prior to assembly				
Circle	4.6.2.2.1	X	3.5.4	X
Level sensitivity	4.6.2.2.2	X	3.5.6.2	X
Reticle line weight	4.6.2.2.3	X	3.5.1.2	X
Post assembly				
Examination	4.6.1	X	3.1 - 3.24	-
Breakdown scales	4.6.2.4	X	3.5.4.5	X
Focus-collimation	4.6.2.5	X	3.5.1	X
Height of standards	4.6.2.9	X	3.5.2	X
Mirror adjustments	4.6.2.10	X	3.5.5	-
Levelment	4.6.2.13.2	X	3.6	-
Theodolite system (horizontal)	4.6.2.3	X	3.22 & 3.23	-
Sand and dust				
Theodolite	4.6.2.15.1	X	3.4.5	-
Carrying case	4.6.2.15.2	X	3.4.5	-
Operation	4.6.2.16	X	3.4 - 3.24	-
Low and high temperature				
Low-temperature operation	4.6.2.17.2	X	3.4.1	-
Operation	4.6.2.16	X	3.4 - 3.24	-
Low-temperature storage	4.6.2.17.3	X	3.4.2	-
Operation	4.6.2.16	X	3.4 - 3.24	-
High-temperature operation	4.6.2.17.4	X	3.4.1	-
Solar radiation	4.6.2.17.6	X	3.4.3	-
Operation	4.6.2.16	X	3.4 - 3.24	-

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TABLE I. Test schedule - continued.

1	2	3	4	5
High-temperature storage	4.6.2.17.5	X	3.4.2	-
Temperature test of wiring	4.6.2.17.7	X	3.12.2.1	-
Operation	4.6.2.16	X	3.4 - 3.24	-
Rainfall				
Theodolite	4.6.2.14.1	X	3.4.4	-
Carrying case	4.6.2.14.2	X	3.4.4	-
Transport case	4.6.2.14.3	X	3.4.4	-
Vibration				
Carrying case	4.6.2.18.1	X	3.4.6	-
Transport case	4.6.2.18.2	X	3.4.6	-
Operation	4.6.2.16	X	3.4 - 3.24	-
Shock				
Shock test of carrying case	4.6.2.19.1	X	3.4.7	-
Shock test of transport case	4.6.2.19.2	X	3.4.7	-
Operation	4.6.2.16	X	3.4 - 3.24	-
Humidity	4.7	X	3.4.8	X
Compass	4.6.2.20	X	3.8	X
Theodolite system (horizontal)	4.6.2.3	X	3.22 & 3.23	-
Telescope				
Effective aperture	4.6.2.6.1	X	3.5.1.1	-
Field of view	4.6.2.6.2	X	3.5.1.1	-
Minimum focus	4.6.2.6.3	X	3.5.1.1	-
Resolution	4.6.2.6.4	X	3.5.1.1	X
Magnification	4.6.2.6.5	X	3.5.1.1	-
Telescope level	4.6.2.6.6	X	3.5.6	X
Reticle lines				
Double	4.6.2.7.1	X	3.5.1.2	-
Single	4.6.2.7.2	X	3.5.1.2	-
Stadia	4.6.2.7.3	X	3.5.1.2	-
Solar circle	4.6.2.7.4	X	3.5.1.2	-
Vertical circle indexing and vertical collimation	4.6.2.8		3.5.6.3	
Altitude level	4.6.2.8.1	X	3.5.6.3	X
Automatic	4.6.2.8.2	X	3.5.6.3	X
Optical plummet				
Focusing	4.6.2.11.1	X	3.5.3	-
Integral with alidade	4.6.2.11.2	X	3.5.3	-
Mounted on tribrach	4.6.2.11.3	X	3.5.3	-
Stadia distance	4.6.2.12	X	3.5.1.2	-
Tribrach				
Base plate nut	4.6.2.13.1	X	3.6	-
Levelment	4.6.2.13.2	X	3.6	-

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4.6.2.1 Conditions of test. Prior to beginning the tests, the contractor shall adjust the theodolite by established methods to operate as specified herein (see 6.6).

4.6.2.2 Prior to assembly.

4.6.2.2.1 Circles. Horizontal and vertical circles shall be tested for accuracy. In carrying out these tests, the circles shall be mounted on a precise turntable with fixed microscopes located so that the comparison can be made of nominally equal angles on different parts of the circle. Any other method of test giving equal results is acceptable. Evidence of a graduation error exceeding 3 seconds or 0.015 mils for horizontal circles or 3 seconds or 0.015 mils for vertical circles shall constitute failure of this test.

4.6.2.2.2 Level sensitivity. The sensitivity of each level vial shall be determined by mounting the level vial on a level trier or by any other method yielding values accurate to within 5 percent. The plane of the graduated vials shall be changed a sufficient amount to cause a change of not less than two graduations in the position of the level bubble. The plane of the ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble as indicated on a scale placed adjacent to the vials. Evidence that the levels do not have the sensitivities specified herein shall constitute failure of this test.

4.6.2.2.3 Reticle line weight. The reticle line weights shall be determined by using a measuring microscope or by using any other method that will yield comparable values. Evidence of the reticle line weights differing from 2.5 to 4.0 microns shall constitute failure of this test.

4.6.2.3 Theodolite system (horizontal). The test shall be reported on a test report form similar to that shown in figure 3. A duplicate of the complete test report form shall be placed within the carrying case of the instrument to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed as part of the test report form:

This test is based on a series of observations on two collimators approximately 36 degrees apart and on a plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested. In making the following observations, each circle reading shall be estimated to the nearest tenth of a graduation of the breakdown scale. This test measures the following parameters of the theodolite:

a. Ability of the instrument to repeat measurements.
b. Average circle graduation error.
c. Collimation error.

Two sets of measurements will be made. Each set will consist of 20 angles (40 measurements) by using all sections of the horizontal circle as follows:

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a. With the telescope in the direct position, the instrument is pointed on the left collimator with a circle reading of approximately 0 degrees and 36 minutes. The circle is read and the seconds are recorded on line 1, column I. The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column I. The telescope is then reversed and pointed on the left collimator, the circle read, and the seconds recorded in line 1, column II (this will be 180 degrees rather than 0 degrees as with the telescope direct). The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column II. The mean of the two readings (columns I and II) is entered in column III. Repeat this technique by changing the position of the horizontal circle to read between 36 and 72 degrees. Advance the circle in this fashion until all parts of the circle are covered during the test.

b. The second set of measurements shall be made identically and shall be recorded in columns IV and V. The mean of these readings shall be in column VI. Taking the difference between the two mean readings from column III, angle 1 shall be determined and shall be written in column VII. Likewise the difference between two mean readings from column VI shall be made and angle 2 shall be determined and written in column VIII. The difference of these two angles represents the pointing and reading error of the test (Δ). These shall be written

in column IX. Column X is Δ^2. From column X, $M_2^2 = \frac{\Sigma\Delta^2}{n}$ shall be calculated, where n represents the total number of readings. In the case of 36 degrees, n = 40; therefore $M_2^2 = \frac{\Sigma\Delta^2}{40} \leq 4 \text{ sec}^2$.

Numbers for larger than 4 sec^2 indicate excessive pointing and reading error. The test setup should be checked and, after correction, two new complete sets of readings shall be made. Excessive values of M_2^2 the second time indicate that the instrument is capable of accurately repeating readings and shall constitute failure of this test.

If M_2^2 is smaller than 4 sec, proceed with calculations as follows:

Means of the two angles are determined from columns VII and VIII and entered in column XI, mean. These are summed and mean is again determined. Residuals are found and entered in column XII (V). Column XII shall be scanned. Maximum plus and minus deviations shall be determined and underlined. Any value in excess of 1.0 sec shall constitute failure of this test. If plus or minus maximum value is not excessive, compute mean (V). Also compute V^2 and enter in column XIII. Zero is entered opposite the first residual from column XII and the algebraic sum of the residuals is entered in column XIV. These values are added algebraically and a mean is determined. This mean is subtracted algebraically from the values of column XIV to give column XV(t). The summation of the column should equal zero. Column XV represents the systematic error of the instrument. Column XVI is t2. Column XVII is the

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difference between column I and column II ($D_1 - R_1$). The sum of this column divided by twice the amount of readings equals the collimation error. In the case of 36 degree collimators, $C = \frac{\Sigma(D_1 - R_1)}{40} \leq$ 12 sec. Any value exceeding 12 sec constitutes failure of this test.

The circle graduation error (t) is computed as follows:

$$M_1^2 = \frac{\Sigma V^2}{n_< - 1} = \frac{\Sigma V^2}{9}$$

$$M_2^2 = \frac{\Sigma\Delta^2}{n_R} = \frac{\Sigma\Delta^2}{40}$$

$$2t^2 = M_1^2 - M_2^2$$

if $\quad M_1^2 - M_2^2 \geq 0$

then $\quad t = \sqrt{\frac{M_1^2 - M_2^2}{2}} \leq 0.08$ min (0.024 *mils*)

Exceeding this value constitutes failure of this test.

if $\quad M_1^2 - M_2^2 < 0$

then compute $\quad (t)m^2 = \frac{\Sigma (t)}{n_<} = \frac{\Sigma (t)}{10}$

$$(t)m = \sqrt{\frac{\Sigma (t)}{10}}$$

Note: (t)m is not the true circle graduation error; it is the average systematic that includes the circle graduation error.

Since $M_2^2 \leq 4$ sec^2 was earlier established, we know that the t $\leq$0.5 sec. This is only true if $M_1^2 - M_2^2 < 0$.

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The number of angles is represented by $n_<$ and n_R is the number of readings.

4.6.2.4 Breakdown scales. The span of the breakdown scales shall be the distance between two adjacent lines of the circles. A difference between any two consecutive circle graduations and the scale graduation lines number 0 and 60 minutes (or 0 and 10 mils) exceeding 0.2 minute or 0.06 mils, as applicable, shall constitute failure of this test.

4.6.2.5 Focus-collimation error. Focus-collimation error shall be determined by observing direct and reverse observations on target at 15 feet (4.6 meters) and infinity (300 feet or 100 meters) and determined as follows:

a. With the telescope in the direct position, sight on the target and read the horizontal circle (e.g., 191 degrees 01.90).
b. Transit the telescope to the reverse position, sight on the target, and read the horizontal circle (e.g., 11 degrees 02.30). Add 180 degrees to this reading (total, 191 degrees 02.30).
c. Subtract the smaller reading from the larger reading (i.e., 91 degrees 02.30 minus 191 degrees 01.90 equals 0.4 minute). The 0.4 minute is twice the collimation error. The error is 0.2 minute.
d. The difference between the error at 15 feet and the error at infinity is the focus-collimation error.

Evidence of a focus-collimation error of more than 0.2 minute (0.06 mils) shall constitute failure of this test.

4.6.2.6 Telescope tests.

4.6.2.6.1 Effective aperture. A scale graduated in millimeters shall be placed in contact with the front of the objective lens. The effective aperture is the diametrical distance of the lens. An aperture of less than 36 mm shall constitute failure of this test.

4.6.2.6.2 Field of view. The field of view shall be measured by pointing the telescope on a collimator or on a target not less than 300 feet (91.4 meters) from the theodolite. The target shall be brought to the intersection of the edge of the field and the horizontal crossline, and the horizontal circle shall be read. The target shall then be brought to the intersection of the horizontal crossline and the opposite edge of the field, and the circle shall again be read. The difference in circle readings is the field of view. The field of view in the vertical plane shall be measured as specific herein, except that the target shall be brought to the intersections of the vertical crossline and the edges of the field, and the readings shall be taken from the vertical circle. Evidence of either field of view measuring less than 1 degree 20 minutes for type I instruments or 24 mils for type II instruments shall constitute failure of this test.

4.6.2.6.3 Minimum focus. The telescope shall be focused on a collimator, the reticle of which has been set to simulate 7 feet (2.2 meters), or the telescope shall be set at minimum focus, and a graduated scale shall be moved

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along the line of sight until the markings are most sharply defined. The distance from this point of the scale measured along the optical axis to the vertical axis of the instrument is the minimum focus. Evidence of the minimum focus being more than 7 feet (2.2 meters) shall constitute failure of this test.

4.6.2.6.4 Resolution. Observations for resolution shall be made within the stadia lines, employing charts contained in NBS Special Publication 374. Resolution of Pattern 56 of the high contrast chart, with the chart placed 77 feet (23.5 meters) from the telescope, indicates a resolution of 4 seconds or better of arc. Inability of the telescope to resolve the pattern shall constitute failure of this test.

4.6.2.6.5 Magnification. The magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light without change of focus; a transparent graduated scale shall be placed in front of the objective lens as near as practicable. A measuring microscope shall be focused on the image of the scale. The magnification of the telescope shall then be determined in diameters as the ratio of a given length of the scale to the length of its image in the exit pupil. When an auto-collimation eyepiece is furnished, the test shall be performed in the same way. Any other method yielding values accurate within 2 percent may be used in lieu of the specified method. Evidence of a magnification of the telescope differing from 28 diameters, ±3 diameters, or of the auto-collimation eyepiece having a magnification less than 18 diameters, shall constitute failure of this test.

4.6.2.6.6 Telescope level. In the reverse position, the telescope shall be leveled by means of vertical indexing system. The telescope level should be in coincidence indicating that the telescope is horizontal. The "peg" method may be used in lieu of the above method. Inability to put the telescope level into coincidence when the line of sight is horizontal shall constitute failure of this test.

4.6.2.7 Reticle lines.

4.6.2.7.1 Double reticle lines. With the theodolite sighted on a collimator, the interval between the double lines shall be measured by means of the horizontal circle. Readings differing from an apparent angle of 40 seconds, +5 or -10 seconds, shall constitute failure of this test.

4.6.2.7.2 Single reticle line. With the theodolite sighted on a collimator, the intervals between the inner edges of the vertical lines and the adjacent edge of the single line shall be observed. These intervals shall be equal. Any notable difference in these intervals shall constitute failure of this test.

4.6.2.7.3 Stadia lines. With the theodolite sighted on a collimator, the length of the horizontal and vertical stadia lines shall be measured by means

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of the horizontal and vertical circles. Any measures less than 5 minutes or more than 7 minutes of arc shall constitute failure of this test.

4.6.2.7.4 Solar circle. With the theodolite sighted and centered on a collimator, the theodolite shall be rotated horizontally 8 minutes with the angle measured by the means of the horizontal circle. Next, the theodolite shall be rotated vertically until an arc of the solar circle is intersected. The vertical angle (V) is then determined by means of the vertical circle.

The solar circle radius (r) can now be calculated $r = \sqrt{64+V^2}$. Calculated

measurements differing from 15 minutes 30 seconds of arc, ±10 seconds, shall constitute failure of this test.

4.6.2.8 Vertical circle indexing and vertical collimation.

4.6.2.8.1 Altitude level. The collimation error of the vertical circle shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, bringing the altitude level to center, and reading the circle. The operation shall be repeated in the reverse position. The sum of the readings should equal 360 degrees or 6400 mils. Any difference includes twice the vertical collimation error. A net error of more than 0.2 minute (0.060 mils) shall constitute failure of this test.

4.6.2.8.2 Automatic. The collimation error of the vertical circle with a self-indexing device shall be determined by sighting in the direct position on a mark with the horizontal crossline of the reticle, and reading the circle. The operation shall be repeated in the reverse position. The sum of the reading should equal 360 degrees or 6400 mils. Any difference includes twice the vertical collimation error. A net error of more than 0.2 minute (0.060 mils) or failure of the device to damp within 5 seconds of time shall constitute failure of this test.

4.6.2.9 Height of standards. A target placed 30 degrees (530 mils) above the horizon of the instrument, and a millimeter scale placed at the same angle below the horizon, shall be used to test the standards. The telescope shall be pointed to the target, then tilted downward onto the scale, and the reading shall be recorded. The telescope shall be transited and another pointing made. Three sets of direct and reversed readings shall be obtained. The angle formed by the target at apex and the two points where the scales has been read is equal to double the angle of tilt of the trunnion axis with respect to the perpendicular of the vertical axis of the instrument. Evidence of any error more than 0.2 minute (0.060 mils) of arc for the sets of readings after correction for the allowable collimation error shall constitute failure of this test.

4.6.2.10 Mirror adjustments. The mirror, or mirrors, shall be adjusted both vertically and horizontally while the breakdown scale is observed. Evidence of any mirror adjustment causing more than 0.1 minute (0.03 mils) variation in the circle readings shall constitute failure of this test.

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4.6.2.11 Optical plummet.

4.6.2.11.1 Focusing. Failure of the optical system of the plummet to produce a clear image throughout the operating range from tripod height to not less than 110 feet shall constitute failure of this test.

4.6.2.11.2 Integral with alidade. The optical plummet shall be tested for alignment with the vertical axis by viewing a target and rotating the alidade 360 degrees. Evidence that the line of sight of the optical plummet is not concentric with the vertical axis of the theodolite on all points of the circle shall constitute failure of this test.

4.6.2.11.3 Mounted on tribrach. The theodolite shall be centered over a ground marker by the plumb bob. The plumb bob shall be removed and the centering checked by means of the optical plummet. This test shall be repeated with the optical plummet over each side of the tripod head. Evidence that the line of sight is not concentric with the vertical axis shall constitute failure of this test.

4.6.2.12 Stadia distance. With the theodolite leveled on a stand or tripod, a graduated rod set at a known distance over 100 feet from the instrument shall be sighted on. The stadia interval shall be read by means of the stadia hairs, first with the rod in the horizontal position and then in the vertical position. This interval shall be multiplied by 100. Evidence of the stadia distance being in error more than ±0.3 percent of the measured distance shall constitute failure of this test.

4.6.2.13 Tribrach.

4.6.2.13.1 Base plate nut. The base plate nut shall be tested for conformance to FED-STD-H28 with certified thread plug gauges. Evidence of any base plate nut not mating with the gauge shall constitute failure of this test.

4.6.2.13.2 Levelment. The theodolite shall be mounted on a tripod or on a trivet which shall be tilted 6 degrees from the horizontal. The instrument shall be leveled by means of the leveling screws. Any evidence that the instrument cannot be leveled shall constitute failure of the test.

4.6.2.14 Rainfall test.

4.6.2.14.1 Rainfall on theodolite. The theodolite, with the eyepiece down, shall be placed in a rain testing chamber not less than 36 inches (91.4 cm) high, 36 inches (91.4 cm) wide, and 36 inches (91.4 cm) deep. The chamber shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches (7.6 cm), ±1 inch (±2.5 cm), per hour. The direction of rainfall shall be varied at a constant rate from directly overhead down to 45 degrees from the vertical. Water temperature shall be 52 °F to 68 °F (11.1 °C to 20 °C) The theodolite shall be within

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±5 °F (±2.8 °C) of the water temperature. The theodolite shall be subjected to this test for 5 minutes. Presence of water in the telescope, optical system, or on bearings shall constitute failure of this test.

4.6.2.14.2 Rainfall on carrying case. The theodolite, in the carrying case, shall be subjected to the test specified in 4.6.2.14.1 for a period of 1 hour. Presence of water in the carrying case, the telescope, or optical system shall constitute failure of this test.

4.6.2.14.3 Rainfall on transport case. The rainfall test of the transport case may be conducted concurrently with the rainfall test of the carrying case, except that the transport case shall be subjected to the rainfall test specified in 4.6.2.14.1 for a period of 1 hour. Presence of water in the transport case, swelling or buckling, loose hardware, or inability to close the case shall constitute failure of this test.

4.6.2.15 Sand and dust test.

4.6.2.15.1 Theodolite. The theodolite shall be placed in a sand and dust testing chamber not less than 36 inches (91.4 cm) high, 36 inches (91.4 cm) wide, and 36 inches (91.4 cm) deep. The sand and dust (140-mesh silica flour (see 6.5) shall pass through the chamber at a velocity of 100 to 500 feet (30.5 to 152.4 meters) per minute. The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at 77 °F, ±5 °F. The relative humidity shall not exceed 30 percent. The theodolite shall be subjected to this test for a period of 15 minutes. The presence of sand or dust in the telescope, optical system, or on bearings, or the presence of lens abrasion, shall constitute failure of this test.

4.6.2.15.2 Carrying case. The carrying case shall be subjected to the test described in 4.6.2.15.1 for a period of 1 hour. Presence of sand or dust in the carrying case shall constitute failure of this test.

4.6.2.16 Operation. The operational test shall be conducted at each time indicated in table I. The theodolite shall be operated and checked as follows to determine its operational capabilities:

a. Manipulate all movable parts and check for freedom of movement and noninterference with other parts in various operating positions throughout their maximum ranges.
b. Fit or attach all accessories to the theodolite and check for functionality as specified.
c. Check the level for free movement to determine if the bubble remains within the graduations and if the bubble ends of the coincidence level remain visible.
d. Rotate the theodolite about its horizontal and vertical axes and check the illumination system for uniform and steady operation.
e. Check microscope focus for uniform and clear focus around each entire circle.

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f. Check parts, components, and accessories for damage.

Evidence that the theodolite, components, or accessories will not function as specified, presence of broken, cracked, or shifted parts on the theodolite or accessories, presence of splinters or cracks, broken, cracked, or loosened hardware, opened joints, shifted padding, or deformation of the cases shall constitute failure of this test.

4.6.2.17 Low-temperature and high-temperature test.

4.6.2.17.1 Test chamber. The low- and high-temperature tests shall be conducted in a test chamber of a size that will house the theodolite mounted on its tripod, the cases, and accessories, and shall permit the inspector to walk around the periphery of the tripod legs in making the examination and test specified herein.

4.6.2.17.2 Low-temperature operation. The theodolite shall be mounted on its tripod and winterization kit covers and shields shall be put in place where applicable. The mounted theodolite shall be covered with the waterproof cover and the theodolite, carrying case, transport case, battery case (batteries excluded), and accessories shall be placed in the test chamber. The temperature in the chamber shall be gradually lowered to -40 °F (-40 °C) and maintained at that temperature until the theodolite is in thermal equilibrium. The batteries (which are not subject to this test) shall then be placed in the battery box and the theodolite shall be connected to the battery box. The theodolite shall then be subjected to the operational test specified in 4.6.2.16. The waterproof cover shall be folded 4 times. Loss of flexibility or cracking of the cover, or inability of the instrument, or of the accessories to operate at -40 °F (-40 °C), shall constitute failure of this test.

4.6.2.17.3 Low-temperature storage. Upon completion of the operational test at low temperature, the theodolite shall be placed in its carrying case which is then placed in the transport case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall be lowered to -50 °F (-45 °C) and maintained at the temperature for 6 hours. The temperature shall then be gradually raised to that of the laboratory and the theodolite dried out and subjected to the operational test specified in 4.6.2.16. Evidence of damage to the theodolite or accessories during storage, or inability of the theodolite to complete the operational test, shall constitute failure of this test.

4.6.2.17.4 High-temperature operation. The theodolite mounted on the tripod, carrying case, transport case, battery case (batteries included), accessories, and waterproof cover shall be gradually raised to 125 °F (51.7 °C) (maintain 85 °F ±2 °F wet bulb) and maintained at this temperature until the theodolite is in thermal equilibrium. The theodolite, cases, and accessories shall then be tested as specified in 4.6.2.16. Evidence of damage to the waterproof cover, or failure of the theodolite or accessories to operate, shall constitute failure of this test.

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4.6.2.17.5 High-temperature storage. Upon completion of the high-temperature operational test, the theodolite shall be placed in its carrying case which is then placed in the transport case. The tripod shall be folded and all accessories stowed for transport. The temperature of the test chamber shall then be raised to 155 °F (68.3 °C) (maintain 8 °F ±2 °F wet bulb) and the temperature maintained for 4 hours. The temperature shall be gradually lowered to that of the laboratory and the theodolite subjected to the operational test specified in 4.6.2.16. Evidence of damage to the theodolite or accessories during storage or inability of the theodolite to complete the operational test shall constitute failure of this test.

4.6.2.17.6 Solar radiation. The theodolite shall be temperature stabilized at 125 °F (51.7 °C) and a source of directed radiant energy simulating solar radiation of 355 Btu/ft^2/hr shall be applied to the theodolite for a period of 4 hours. At the end of this period, the theodolite shall be tested as specified in 4.6.2.16. In addition, the theodolite shall be observed for peeled, cracked, or blistered paint. Presence of any of the defects specified in 4.6.2.16 or of peeled, cracked, or blistered paint shall constitute failure of this test.

4.6.2.17.7 Temperature test of the wiring. The temperature test of the wiring shall be conducted by using the battery and hand lamp cords. Temperature tests of the wiring may be conducted concurrently with the temperature tests of the theodolite.

4.6.2.17.7.1 Low-temperature operation test of wiring. The battery and hand lamp cords shall be in the battery and lighting accessory case when placed in the temperature control chamber. The two cords shall remain in the test chamber during the low-temperature operation tests of the theodolite. At the end of the test, and without being removed from the chamber, the two cords shall be bent around a 3/8-inch mandrel at a rate of 12 turns per minute. The two cords shall then be removed from the chamber and examined for cracks and flaws under a glass having a magnification of 3 diameters. Evidence of damage shall constitute failure of the test.

4.6.2.18 Vibration. The theodolite shall be subjected to vibration testing as specified herein.

4.6.2.18.1 Carrying case removed from the transport case. The theodolite in the carrying case shall be rigidly mounted to the vibration table and subjected to two sweeps of sinusoidal cycling vibration in each of three mutually perpendicular axes. Each sweep shall be 5 to 55 to 5 Hertz (Hz) and traversed in 15 minutes. The vibration amplitude shall be 0.10 inch double amplitude from 5 to 17 Hz and 1.5g from 17 to 55 Hz. Upon completion of this test, the theodolite shall be mounted on its tripod and tested for operation as specified in 4.6.2.16. Evidence of damage to the theodolite or the case shall constitute failure of the test.

4.6.2.18.2 Transport case. The theodolite in the transport case shall be tested in accordance with MIL-STD-810, procedure X, table 514.2-VII, figure

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514.2-7, curve AX except that only five sweeps of vibration cycling shall be performed for a total of 1 hour of vibration in each axis. Upon completion of this test, the theodolite shall be mounted on its tripod and tested for operation as specified in 4.6.2.16. Evidence of damage to the theodolite or the case shall constitute failure of the test.

4.6.2.19 Shock.

4.6.2.19.1 Shock test of carrying case. The theodolite in the carrying case shall be flat-dropped in an upright position from a height of not less than 6 inches (15.2 cm) onto a level wood floor. The theodolite shall then be subjected to the operation test specified in 4.6.2.16. The theodolite in its carrying case shall then be placed in the transport case and, after being secured for shipment, shall be dropped from a height of 12 inches (30.5 cm) so that one bottom edge of the transport case strikes a level concrete floor when the bottom of the transport case is inclined 15 to 20 degrees from the horizontal. The theodolite shall again be tested as specified in 4.6.2.16. Evidence of damage to the theodolite or to the cases shall constitute failure of this test.

4.6.2.19.2 Shock test of transport case. A dummy load in the configuration of the carrying case, and equal in weight to the theodolite and the carrying case, shall be placed in the transport case. The transport case shall be closed and secured as for shipment and subjected to eight drops, one on each bottom edge and one on each corner, onto a horizontal concrete floor from a height of not less than 42 inches (106.7 cm). The height of drop shall be measured from the concrete floor to the nearest edge of corner of the transport case when suspended prior to each drop. Evidence of damage to the case as specified in 4.6.2.16 shall constitute failure of this test.

4.6.2.20 Compass. With the compass mounted on the theodolite, the theodolite mounted on its tripod shall be set up and plumbed over a fixed point that is located on a line of known azimuth. A target shall be established for reference in the north and south line. The horizontal circle shall be oriented so that 0 degree (0 mils) is read on the horizontal circle when the telescope is in the direct position and is sighted on the target. Readings shall then be taken at: 0 degrees (0 mils), 45 degrees (800 mils), 90 degrees (1600 mils), 135 degrees (2400 mils), 180 degrees (3200 mils), 225 degrees (4000 mils), 270 degrees (4800 mils), and 315 degrees (5600 mils) by means of the horizontal circle and the compass dial. The readings taken on the compass dial shall be compared with the readings taken on the horizontal circle. If readings taken on the compass dial do not differ more than ±30 minutes (9 mils) from readings taken on the horizontal circle, the upper rotatable part of the theodolite shall be considered free from magnetic influence. The theodolite shall be reset over the same point with the tripod and lower stationary part of the theodolite rated through approximately 90 degrees. The horizontal circle shall be oriented so that 0 degrees (0 mils) is read on the horizontal circle when the telescope is in the direct position and is sighted on the target. Readings shall be taken at: 45 degrees

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(800 mils), 225 degrees (4000 mils), 270 degrees (4800 mils), 315 degrees (5600 mils), and 360 degrees (6400 mils) by means of the horizontal circle and the compass dial. The readings taken on the compass dial shall be compared with the readings taken on the horizontal circle. If readings taken on the compass dial do not differ more than ±30 minutes (9 mils) from readings taken on the horizontal circle, the lower part of the theodolite shall be considered free from magnetic influence. Any readings on the compass dial which is more than ±30 minutes (9 mils) different from the corresponding readings on the horizontal circle, or, after compass readings have been corrected for declination, failure to produce a true azimuth within ±30 minutes (9 mils), or failure of the compass dial to rotate uniformly without sticking shall constitute failure of this test.

4.7 Humidity. Place the theodolite with power off, mounted in the operating position with all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2, procedure II. The theodolite shall be inspected and tested as specified in 4.6.2.16. Evidence of damage shall constitute failure of this test.

4.8 Inspection of packaging. The preservation, packing, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete theodolite with all components and accessories provided, shall be preserved and packed in accordance with MIL-S-3534. The level of preservation and packing shall be level A, B, or C as specified (see 6.2). Marking for levels A, B and C shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I theodolite are primarily intended for engineer construction and topographic third- and forth-order survey work (see 6.7). Type II theodolites are primarily intended for artillery and missile survey.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of theodolite required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of first article theodolite (see 3.2).

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e. When the Government will conduct any or all of the first article theodolite examination and tests. When the Government will conduct some but not all of the first article theodolite examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When an autocollimation eyepiece is required (see 3.5.1.3.4).
g. When an altitude level for the vertical circle index is required (see 3.5.6.3).
h. When a compass is not required (see 3.8).
i. When a hand lamp is required (see 3.12.2.2)
j. When battery and lighting accessory cable is required (see 3.12.2.4).
k. When a tripod is required (see 3.13).
l. Color required (see 3.17 and 3.21).
m. When a knapsack is required (see 3.17).
n. When tools and accessories are required (see 3.18).
o. Grease required if other than specified (see 3.19).
p. Level of preservation and packing required (see 5.1).

6.3 First article. When a first article inspection is required, the items(s) should be a preproduction model. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Information figures. Figure 1 shows a reticle pattern and figure 2 shows a type of stub, both of which have been found acceptable. However, the figures have been included for illustration only and are not intended to preclude the furnishing of other means of attaching the alidade to the tribrach or of other reticle patterns which conform to this specification.

6.5 Sand and dust. The sand and dust to be used in the sand and dust test is commercially known as "140-mesh silica flour". Sand and dust, 140-mesh silica flour produced by the Fenton Foundry Supply Company, Dayton, OH 45403, and Ottawa Silica Company, Ottawa, IL 61350, or equal will be satisfactory.

6.6 Condition for inspection. The contracting officer should require the contractor to certify that the equipment is ready for examination and testing as specified in table I.

6.7 Accuracy. Information regarding third- and fourth-order accuracy will be found in Department of Commerce Publication: "Classification, Standards of

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Accuracy and General Specifications of Geodetic Control Surveys" and "Specifications to Support Classification, Standards of Accuracy and General Specifications of Geodetic Control Surveys."

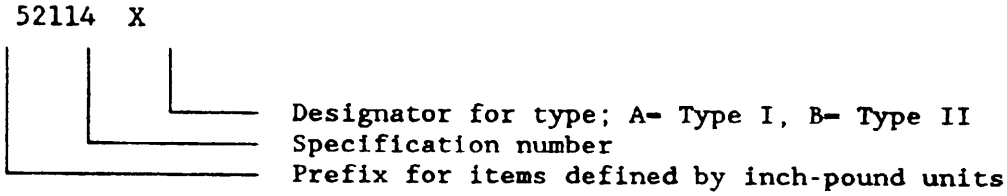
6.8 Provisioning. The contracting officer should include provisioning requirements for such repair parts and maintenance tools as necessary (including any special tools, and instructions) on shipment of theodolites.

6.9 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be listed on a separate Contract Data Requirements List (DD form 1423), which is included as an exhibit to the contract. The Technical Manual must be required under separate contract line item in the contract.

6.10 Subject term (key word) listing.

a. Alidade
b. Direction
c. Integral illumination system
d. Missile survey
e. Optical-reading

6.11 Part or identifying number (PIN). The PIN to be used for theodolites acquired to this specification are created as follows:



6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Review activity:
DLA - GS

User activities:
Army - AR
Navy - MC

Preparing activity:
Army - ME

Project 6675-0938

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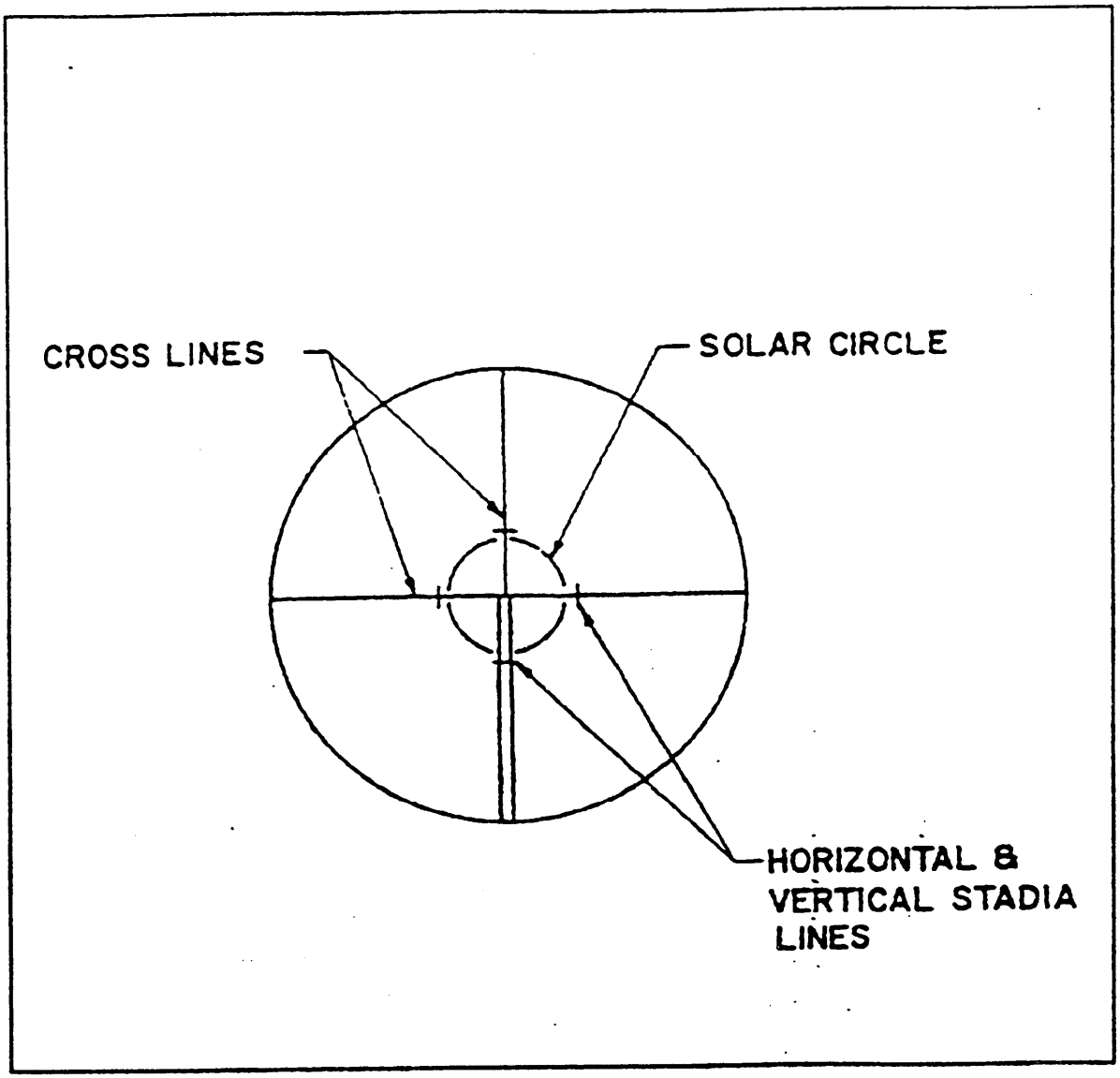


FIGURE 1. Reticle Pattern for Theodolite.

X-3104A

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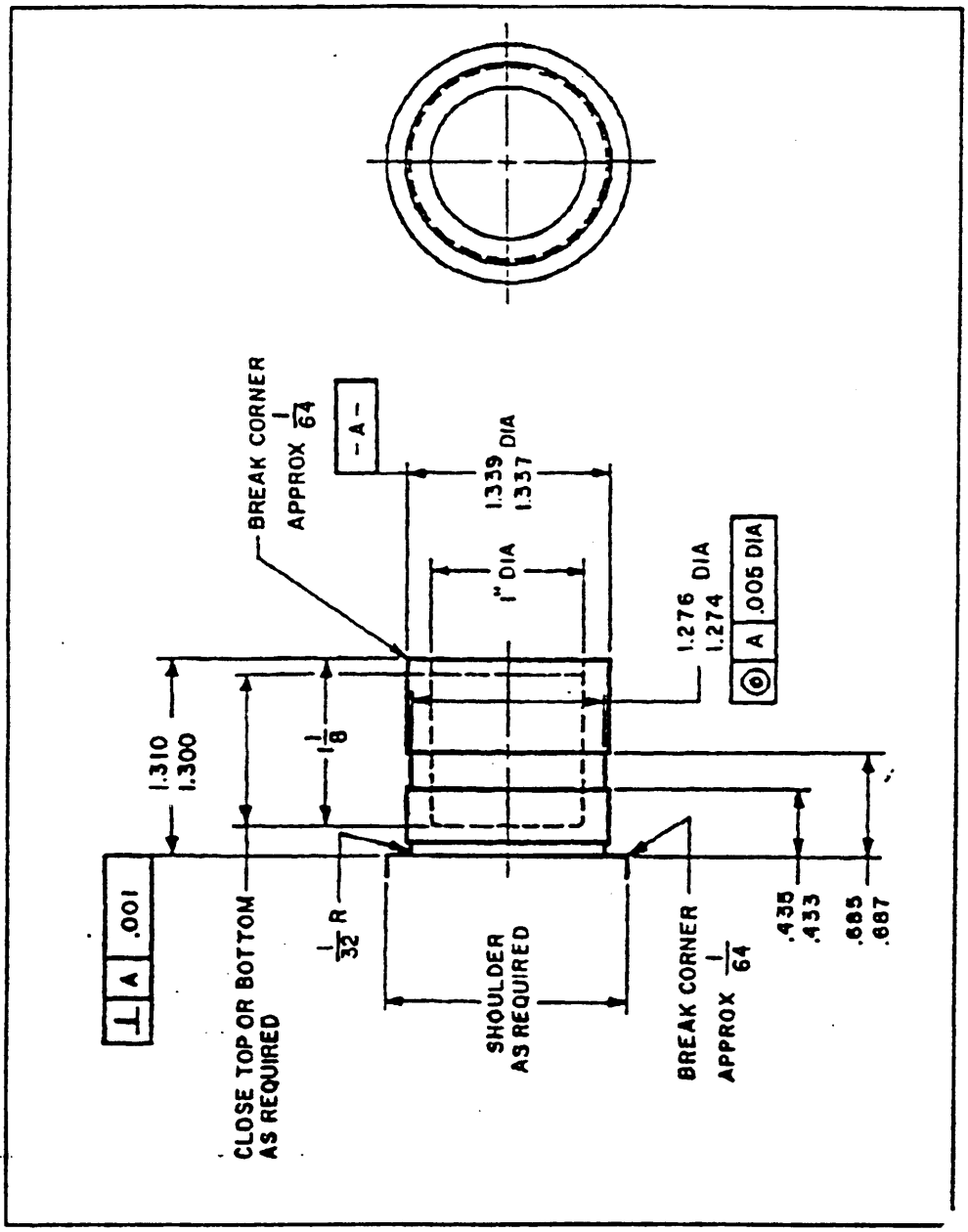


FIGURE 2. Theodolite Stud.

CX-1563

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Column		I	II		III	IV	V		VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII
Pointing		Mils Min B_1 Sec	Mils Min B_2 Sec		$\frac{B_1+B_2}{2}$	Mils Min D_1 Sec	Mils Min D_2 Sec		$\frac{D_1+D_2}{2}$	<1 (B-A)	<2 (B-C)	(<1 - <2)	t	$\frac{<1 + <2}{2}$	V (XI - M_1)	V^2	R ($V_1+V_2+V_3$)	(t) (XIV - M)	$(t)^2$	D_1-A_1
Deg	Mil																			
0	0			A				C												
36	640			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
36	640			A				C												
72	1280			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
72	1280			A				C												
108	1920			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
108	1920			A				C												
144	2560			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
144	2560			A				C												
180	3200			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
180	3200			A				C												
216	3840			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
216	3840			A				C												
252	4480			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
252	4480			A				C												
288	5120			B				B		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
288	5120			A				C												
324	5720			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
324	5720			A				C												
0	0			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
														I M -	I V_2 -		I R -		I C -	
												$M_2^2 = \frac{I_2^2}{40}$ -		$M_1 = \frac{I_R}{10}$ -	$M_1^2 = \frac{I_{V_2}^2}{9}$ -		$M_1 = \frac{I_R}{10}$ -		$C = \frac{I_C}{40}$ -	

$2t^2 = M_1^2 - M_2^2$ if $M_1^2 - M_2^2 \geq 0$ then $t = \pm\sqrt{\frac{M_1^2 - M_2^2}{2}}$

Mfgr _____ Date _____

Ser. No. _____

Order _____

Tested by _____

Checked by _____

FIGURE 3. Data Sheet for Theodolite System Test (horizontal). X-3034

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